FPV DRONE TRACKING COMPETITION
Pre-Design Report

TEKNOFEST
AEROSPACE AND TECHNOLOGY FESTIVAL

Contents

1. Definitions and Abbreviations 3
2. Project Summary 3
3. Organization Summary 4
3.1 Team Organization 4
3.2 Timeline and Budge 4
4. PROJECT PRE-DESIGN 4
4.1 Electronic Design 5
4.1.1 Propulsion System 5
4.1.2 Power System 5
4.1.3 Flight Control System 5
4.1.4 Image Processing System 6
4.2 Software Design 6
4.2.1 ROS2 Node Architecture 6
4.2.2 Detection Software 7
4.2.3 Tracking Software 7
4.2.4 Reacquisition 8
4.2.5 Control Software: Image-Based Visual Servoing 8
4.2.6 State Machine 8
4.3 Mechanical Design 8
4.3.1 Fuselage Construction Material 8
4.3.2 Fuselage Design and Manufacturing 8
4.3.3 Basic Calculations 9
5. Method 9
6. References 10

1. Definitions and Abbreviations

Abbreviation	Definition
FC	Flight Controller
ESC	Electronic Speed Controller
UBEC	Universal Battery Elimination Circuit
KV	Motor velocity constant (RPM per Volt)
GCS	Ground Control Station
IMU	Inertial Measurement Unit
MAVLink	Micro Air Vehicle communication protocol
PDB	Power Distribution Board
ROS2	Robot Operating System 2
IBVS	Image-Based Visual Servoing
PID	Proportional-Integral-Derivative (controller)
FSM	Finite State Machine
YOLO	You Only Look Once (object detector)
mAP	Mean Average Precision
IoU	Intersection over Union
NPU	Neural Processing Unit
UART	Universal Asynchronous Receiver-Transmitter
AUW	All-Up Weight
CF	Carbon Fibre
VTX	Video Transmitter

2. Project Summary

This project develops a fully autonomous FPV drone system capable of detecting, locking onto, and continuously tracking a target drone through a sequence of eleven

predefined maneuvers inside a competition area.relying entirely on onboard vision-based perception.

Target detection is performed using a YOLOv8n model trained on a multi-source dataset of drone imagery. Tracking is maintained frame-to-frame using a fusion of Lucas-Kanade sparse optical flow and SiamRPN++, with periodic YOLO corrections to prevent drift accumulation. When target loss is detected, a three-stage reacquisition cascade employing Kalman prediction, progressive region expansion, and appearance filtering recovers the target with minimal latency. All perception outputs feed into an (IBVS) cascade PID controller, which commands the flight controller directly in velocity space without requiring any 3D world reconstruction or depth estimation.

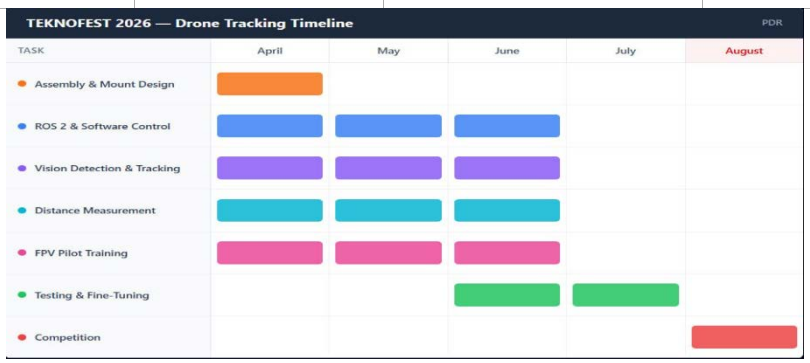
3. Organization Summary

3.1 Team Organization

No.	Name - Surname	Role in Team	class	EDUCATIONAL LEVEL	Member Role
1	Abdalrhman Qasomeh	Captain	Mechanical Design Engineer	Bachelor	captain
2	Hamzeh abo alzahab	Detection & pilot	ITE	Bachelor	Member
3	Mustafa hazzaa	Tracking	ITE	Bachelor	Member
4	Abdalrhman rasheed	Control & Integration Lead	Computer automation Engineering	Bachelor	Member
5	Abdulrahman Alkhani	Electronics & Power & Mechanical Design	Computer automation Engineering	Bachelor	Member

3.2 Timeline and Budge

Our budget is about 800$ with spare parts for errors



4. PROJECT PRE-DESIGN

4.1 Electronic Design

our electronic architecture is divided into four tightly integrated subsystems: propulsion, power management, flight control, and video transmission. Each component has been selected to ensure compatibility, reliability, and compliance with competition requirements.

4.1.1 Propulsion System

The propulsion system drives all maneuverability and tracking performance of the drone. The selected components are:

Motors (x4)	Angel A2807(1700KV)Brushles
Propellers (x4)	HQPROP 7X4X3 7040 Three-Blade
ESC	YSIDO 45A 4-in-1 BLHeli_S ESC

The Angel A2807 1700KV motors are rated for 4S LiPo operation, providing adequate thrust-to-weight ratio for a 7-inch platform. The 1700KV rating is optimal for 7-inch three-blade propellers, balancing efficiency and response time which is critical for stable image acquisition during target tracking.
The YSIDO 45A 4-in-1 BLHeli_S ESC integrates all four motor controllers into a single compact board, minimizing wiring complexity and weight. The 45A continuous current rating provides a comfortable safety margin over the expected peak draw of the A2807 motors on a 4S supply.

4.1.2 Power System

The 4S LiPo (14.8 V) provides up to 176 A (80C), preventing voltage sag during simultaneous high-throttle and compute loads. A 200 A ANL fuse on the battery positive protects the full power path. System power is switched through a 200 A relay controlled by a low-current toggle on the 5 V rail.
The Pixracer R15 receives regulated 5 V from the YSIDO ESC's power distribution, which also powers the receiver and telemetry. The Raspberry Pi 5 is supplied by the HENGE 6A UBEC.

4.1.3 Flight Control System

Flight Controller	Pixracer R15
RC Receiver	JHEMCU ExpressLRS RX24T 2.4GHz ELRS
RC Transmitter	RadioMaster Pocket Edge TX 2.4GHz
Telemetry	SiK Telemetry Radio V3

The flight control system uses the Pixracer R15, which provides integrated IMU, barometer, and magnetometer sensing with full PX4 support for autonomous operation. It sends motor control signals to the YSIDO 4-in-1 ESC via the DShot600 digital protocol and operates in PX4 OFFBOARD mode during autonomous tracking, requiring velocity setpoints from the companion computer at ≥2 Hz. Manual override

is ensured through the JHEMCU ELRS RX24T 2.4 GHz receiver with sub-5 ms latency. System failsafes include zero-velocity output on target loss, PX4 OFFBOARD timeout for controlled hold, and immediate RC override via mode switch.

4.1.4 Image Processing System

FPV Camera (Pilot View)	RunCam Phoenix 2 SE V2
Tracking Camera	Pi Camera UltraWide 60fps FHD
Video Transmitter	RUSH Solo Tank 5.8GHz VTX (CNC Shell)
FPV Goggles	Eachine EV800D 5.8GHz 40CH 5-inch

The RunCam Phoenix 2 SE V2 provides the pilot-view FPV feed, delivering low-latency analog video to the RUSH Solo Tank VTX, which transmits the signal to the Eachine EV800D goggles for real-time situational awareness. The AI-processing camera is the Raspberry Pi Camera Module 3 Wide, we choose it for its field of view and 60 fps capability, enabling reliable target tracking and direct integration with the Raspberry Pi 5. The RUSH Solo Tank VTX is used for its stable analog transmission and proven operational reliability.

4.2 Software Design

The complete software system is structured as a directed graph of modular ROS2 nodes. Each node holds a single responsibility This modularity provides fault isolation. The system runs on Ubuntu 22.04 (ROS2 Humble).

4.2.1 ROS2 Node Architecture

Table 2: ROS2 Node Pipeline Summary

Node	Input Topic	Output Topic	Responsibility
camera_node	Hardware (CSI)	/camera/image_raw	Capture and publish raw frames
detection_node	/camera/image_raw	/yolo/detections	YOLOv8n inference,
tracking_node	/yolo/ detections+ /camera/image_raw	/tracker/offset + /tracker/status	Optical flow, SiamRPN, Kalman fusion, offset
state_machine	/tracker/status	/drone/state	FSM state transitions,

			reacquisition trigger
control_node	/tracker/offset + /drone/state	/mavros/.../cmd_vel	Cascade PID IBVS, velocity setpoints
mavros2_node	/mavros/setpoint_velocity/cmd_vel	MAVLink UART to PX4	ROS2 to MAVLink translation

The control node publishes velocity setpoints to **MAVROS**, which converts them into MAVLink messages and transmits them to the Pixracer flight controller running PX4 in OFFBOARD mode.
Setpoints are published continuously to maintain OFFBOARD control; if tracking is lost, zero-velocity commands are sent to ensure safe behavior.

4.2.2 Detection Software

YOLOv8n is selected for its fast, single-pass, anchor-free architecture, enabling real-time drone detection on the Hailo-8L and maintaining robustness to. Training data combines open-source datasets (Anti-UAV, MAV-VID), and other methods Standard augmentations are applied, and the model will be fine-tuned later.

Table 3: Detection Evaluation Metrics

Metric	Formula	Role
mAP@0.5	Area under P-R curve	(≥0.85 required)
Recall	TP / (TP + FN)	Minimises reacquisition events
Precision	TP / (TP + FP)	Reduces false tracker lock-ons
F1-Score	2 × P × R / (P + R)	Balanced single-value indicator for dataset comparison

The detection model must achieve mAP@0.5 ≥ 0.85 on the validation set before hardware integration proceeds. Recall is prioritised as the primary operational metric because each missed detection triggers the reacquisition module

4.2.3 Tracking Software

Our system employs a priority-weighted fusion strategy optimized for edge hardware (Raspberry Pi 5) to ensure drift-free, real-time target acquisition.

Lucas-Kanade (LK) Flow: High-frequency backbone (<2ms); maintains bounding box estimates between neural network cycles.
SiamRPN++ (15-25 FPS): Primary tracker; uses template cross-correlation to handle scale changes and partial occlusions.
YOLO (Periodic): Performs absolute drift correction and full-frame re-detection after target loss.
Fusion Logic & Kalman Filter: A 6D filter [x, y, w, h, vx, vy] smooths output. Coordinates are resolved by confidence: YOLO (>0.8) overrides SiamRPN++ (>0.6), with LK as the baseline fallback.

4.2.4 Reacquisition

Target loss is declared when tracking confidence falls below 0.3 for five consecutive frames, triggering the TARGET_LOST state. Recovery is performed using a three-stage cascade: first, Kalman prediction restricts the search to a region of interest for fast recovery; second, the search area is progressively expanded to full-frame coverage; and third, candidate detections are validated using appearance features to reject false positives. A confirmation step requiring three consecutive detections ensures stable re-entry into tracking.

4.2.5 Control Software: Image-Based Visual Servoing

Control is implemented using (IBVS), where the drone minimizes the error between the target position and the image center. This approach avoids the need for depth estimation or global positioning, making it suitable for GNSS-denied environments. A cascade PID structure is used. The outer loop converts pixel error into desired velocity, while the inner loop tracks this velocity using feedback from the flight controller. Horizontal error controls lateral motion and yaw, vertical error controls forward motion, and bounding box size is used to regulate altitude.

4.2.6 State Machine

System behavior is governed by a finite state machine with the following states: IDLE (initialization), SEARCHING (target scanning), TRACKING (normal operation), UNCERTAIN (reduced confidence), TARGET_LOST (tracking failure), REACQUIRING (target recovery), and LANDING (mission completion). Each state defines specific control actions and ensures safe transitions between tracking and recovery modes.

4.3 Mechanical Design

The mechanical design is based on the Mark4 7-inch frame, selected for its structural rigidity, reliability, and compatibility with standard FPV components. The design meets mission requirements for high maneuverability, vibration resistance, and stable sensor mounting during dynamic tracking.

4.3.1 Fuselage Construction Material

The frame is constructed from carbon fiber, providing a high stiffness-to-weight ratio. The 5 mm arms minimize flex under thrust loads, reducing vibration and improving flight stability. The 3 mm central plates form a rigid structure for mounting the battery and onboard electronics.

4.3.2 Fuselage Design and Manufacturing

The frame uses a true-X configuration, ensuring symmetric mass distribution and simplified control tuning. The 295 mm motor-to-motor distance supports 7-inch propellers, suitable for efficient and stable flight. The tracking camera is mounted on a fixed, slightly upward-tilted mount to compensate for forward pitch during motion, maintaining the target within the field of view. Rubber dampers are used to reduce vibration and preserve image quality.
Custom mounts (camera and antenna) are manufactured using 3D printing with PETG or ASA materials, selected for their durability and resistance to environmental conditions.

4.3.3 Basic Calculations

The estimated all-up weight is approximately 850 g. Each motor produces up to ~1990 g of thrust, resulting in a total thrust of ~7960 g and a thrust-to-weight ratio of ~9.4:1 (~4.2:1 during tracking), exceeding the minimum requirement for agile flight. The estimated hover current is 20–25 A, with a flight time of 5–7 minutes under dynamic conditions. Which meets the requirements.

5. Method

The system is a fully autonomous, vision-based tracking platform operating without GNSS, using an integrated perception, tracking, and control pipeline.

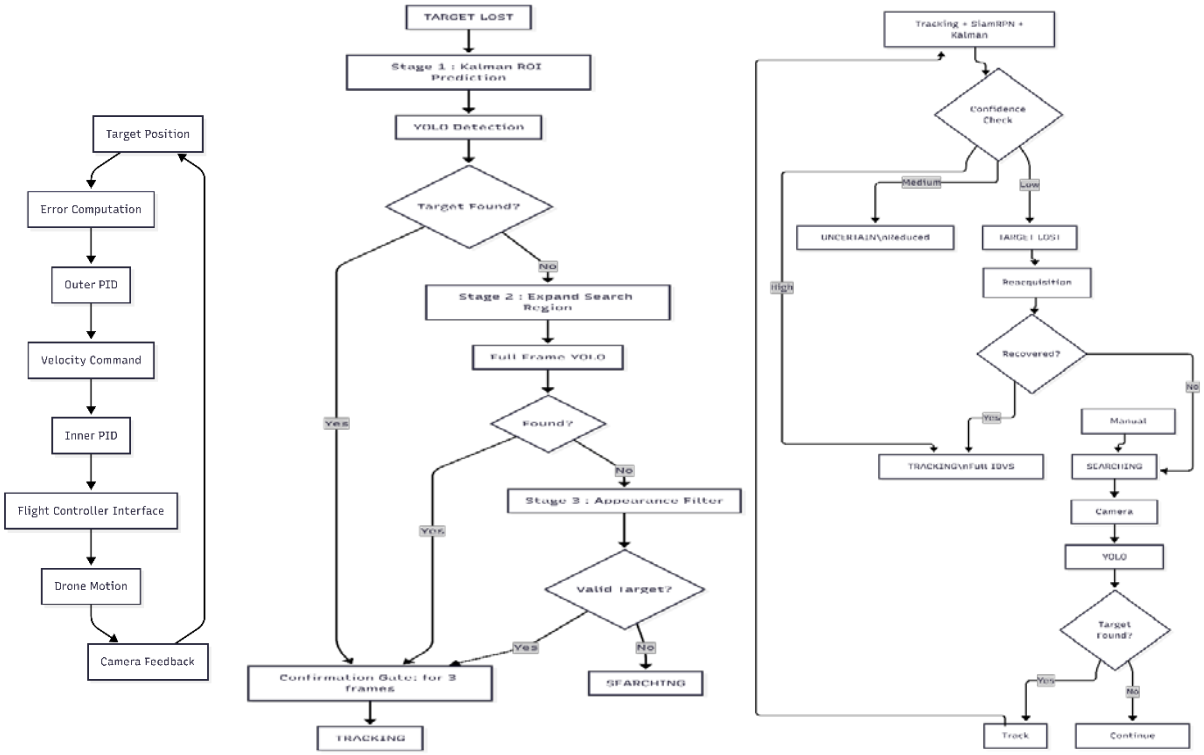
Two cameras each serving its on purpose (FPS or Quality) provides real-time video processed on a Raspberry Pi 5 with AI acceleration. Target detection is performed using YOLOv8n, chosen for its real-time performance and robustness to varying drone shapes.

Tracking is maintained through a hybrid approach combining Lucas-Kanade optical flow for high-frequency updates, SiamRPN++ for robustness to motion and occlusion, and periodic YOLO corrections to eliminate drift. A Kalman filter fuses these inputs into a stable, predictive state estimate.

Upon target loss, a three-stage reacquisition strategy is applied: Kalman-based region prediction, progressive search expansion, and appearance-based validation to ensure reliable recovery.

Control is implemented using IBVS, where image-space error is converted into velocity commands via a cascade PID controller and sent to the flight controller through MAVLink in OFFBOARD mode, removing the need for depth estimation.

The system is structured as a modular ROS2 architecture governed by a finite state machine, ensuring safe transitions between search, tracking, and recovery states.



6. References

[1] ArduPilot Development Team, Pixracer Flight Controller Documentation, not explicitly stated,4-2026, https://ardupilot.org/copter/docs/common-pixracer-overview.html

[2] Raspberry Pi Ltd., Camera Hardware Documentation, not explicitly stated, April-2026, https://www.raspberrypi.com/documentation/accessories/camera.html

[3] Jocher, G., Ultralytics YOLOv8, 2023, April-2026, https://github.com/ultralytics/ultralytics

[4] SenseTime Research, PySOT Visual Tracking Toolbox, not explicitly stated, April-2026, https://github.com/STVIR/pysot

[5] OpenCV Team, OpenCV Library Documentation, not explicitly stated, April-2026, https://docs.opencv.org/

[6] Welch, G., Bishop, G., An Introduction to the Kalman Filter, 2006, April-2026, https://www.cs.unc.edu/~welch/media/pdf/kalman_intro.pdf

[7] MAVLink Development Team, MAVLink Protocol Documentation, not explicitly stated, April-2026, https://mavlink.io/en/messages/common.html

[8] Open Robotics, ROS2 Documentation, not explicitly stated, April-2026, https://docs.ros.org/en/humble/index.html

[11] Hailo AI, Hailo-8L AI Accelerator Datasheet, not explicitly stated, April-2026, https://hailo.ai/products/hailo-8l/a